



Real-Time Thermospheric Density Forecasting with a Metric-Aligned BiGRU-Attention Network



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Abstract

Atmospheric density forecasts are essential for collision-risk mitigation in the increasingly congested Low-Earth Orbit. We introduce a fully data-driven pipeline that converts a 60-day history of hourly OMNI2 solar-wind indices and orbital state vectors into a 72h orbit-averaged density forecast at 10min cadence. The predictor is a lightweight three-layer bidirectional GRU equipped with additive attention; training uses a “metric-aligned” loss — an exponentially time-weighted mean-squared error whose decay kernel is identical to the OD-RMSE skill score used by the STORM-AI leaderboard, thereby optimizing the model directly for the problem statement criterion. Per-sequence density normalization and multi-scale lag features further boost storm-time fidelity. Trained on OMNI2 satellite data, the network attains a normalized model score of 0.709 on the STORM-AI Phase 1 leaderboard, placing 5th out of 18 participants. Future work will fuse physics priors and uncertainty quantification, but the current prototype already offers a real-time, drop-in alternative to empirical climatologies. ^a

^aAll the code can be found in Github: <https://github.com/atillasaadat/stormai-mit-competition>

Introduction

Thermospheric mass density is the dominant uncertainty in Low-Earth Orbit (LEO) drag modeling; during geomagnetic storms it can spike by an order of magnitude, compromising conjunction assessment and station-keeping for the $\sim 5,500$ operational satellites and the far larger constellations now planned [1]. First-principles general-circulation models (e.g. TIE-GCM, GITM) resolve fine-scale dynamics but demand super-computing resources and are too slow for operational multi-day forecasts [2]. Empirical climatologies such as NRLMSISE-00 and JB2008 run in milliseconds yet miss post-storm cooling, composition changes, and routinely has prediction errors of $> 40\%$ [3].

Data-driven machine-learning methods now offer a computationally efficient middle ground between costly first-principles GCMs and limited empirical climatologies. Reduced-order-model surrogates retain key TIE-GCM modes with $\sim 5\%$ 24-h error yet lose fidelity under disturbed dynamics [4]. Deep networks trained on physics residuals shorten one-step errors by two orders of magnitude [5]; transformers outperform linear ROMs across solar cycles [1]. Neural emulators of HASDM [6], MBiLE ensembles [7], orbit-specific

Bi-LSTM/transformer blends [8], and probabilistic PCA-RNN hybrids [9] all approach physics-model accuracy while running faster by a factor of $10^2 - 10^4$.

Motivated by these advances, we adopt a fully data-driven strategy: a lightweight Bidirectional GRU with additive attention directly maps 60 days of solar-wind history and orbital context to a 72-h forecast of orbit-averaged density, eschewing intermediate surrogates. The model meets the STORM-AI challenge horizon while retaining real-time throughput, and its design choices—metric-aligned loss, per-sequence normalization, and multi-scale lag features—target storm-time fidelity and cross-satellite generalization.

Approach Definition

We adopt a fully data-driven, end-to-end pipeline. A 60-day slice of drivers feeds the model; each hourly row fuses (i) OMNI2 solar-wind & geomagnetic indices, (ii) Keplerian orbit parameters plus derived latitude/longitude/altitude, and (iii) sinusoidal encodings of longitude, day-of-year, and sidereal time. To expose multiple temporal scales we append six explicit lags $\{1, 2, 3, 4, 6, 12\}$ h and a 3h moving-average, a total of 181 features per step. Prior to modeling, each sequence is normalized by a file-level prior ρ_0 (the mean density observed below 1 kg m^{-3}) so that the network learns relative fluctuations that transfer across altitude regimes. A quantile transformer then maps every feature to an approximate $\mathcal{N}(0, 1)$ distribution, mitigating the heavy tails of proton flux and K_p .

The architecture is a lightweight three-layer Bidirectional Gated Recurrent Unit (BiGRU) ($h = 384$ per direction) that produces hidden states $\mathbf{H} \in \mathbb{R}^{1440 \times 768}$. Additive attention assigns a score e_t to each hour and forms a context vector, $\mathbf{z}$ with α_t , defined in Equations (6) and (5). This automatically up-weights storm-time segments that dominate density variability. A LayerNorm-GELU-Dropout block reduces $\mathbf{z}$ to 384 dimensions; a final linear layer outputs a vector of 432 $\log_{10}$ -scaled residuals, which are rescaled to physical atmospheric density units at inference.

Training minimizes a time-weighted MSE whose exponential weights mirror the STORM-AI leaderboard metric, ensuring fidelity at both short and long lead times. Optimization uses AdamW ($\eta_0 = 3 \times 10^{-4}$, weight decay 10^{-4}), gradient clipping (1.0), mixed-precision AMP, and early stopping (patience 24). This design converges in ~ 4 h on a single RTX 3090 Ti.

Compared with physics-informed hybrids that regress on TIE-GCM modes or MSIS residuals, our model operates directly on the observable and eliminates expensive forward models, echoing the fully empirical philosophy of [6]. The BiGRU-attention encoder preserves temporal order while using an order of magnitude fewer parameters than transformer baselines [1], and the sequence-wise normalizer with metric-aligned loss notably improves cross-satellite generalization—a limitation noted in [7]. Together, these innovations deliver state-of-the-art accuracy without sacrificing operational throughput or reproducibility.

Data & Pre-processing

The study combines three complementary data streams, each keyed by a five-digit *File ID*. (i) Space-weather drivers: hourly OMNI2 measurements (solar wind plasma, interplanetary magnetic field, and geomagnetic indices) span 1995-2023 with $< 3\%$ missing entries. (ii) Initial Orbit State: single-epoch Keplerian elements and geodetic coordinates supplied by the STORM-AI dev-kit capture satellite altitude and local-time geometry to initialize the satellite orbit to predict along. (iii) Targets: ten-minute orbit-averaged densities derived from on-board accelerometers of ESAs Swarm and GRACE missions as truth data. After a one-day spin-up period, the intersection of the three streams yields 6915 complete triplets; any file containing fewer than 432 valid density rows (three days at 10min cadence) is discarded, and any file with more is truncated to the first 432 density rows.

OMNI2 encodes instrument dropouts with sentinel values $\{99999.99, 9999999, -1\}$. These are mapped to NAN, the series is resampled to a strict 1-hour grid, brief gaps (< 3 h) are linearly interpolated, and longer outages are forward/backward filled. Rolling means of three hours are aligned with the hourly index to capture mesoscale variability, and six explicit lags $\{1, 2, 3, 4, 6, 12\}$ h are added at resolutions of 1h and

3h, resulting in 168 dynamic channels with deterministic column order.

Orbital state records undergo independent validation. If any of the initial orbit states' latitude, longitude, altitude violates physical bounds, geodetic coordinates are re-derived from Keplerian elements via the classical ECI $\rightarrow$ ECEF rotation

$$\mathbf{r}_{\text{ECI}} = \mathbf{R}_3(-\Omega) \mathbf{R}_1(-i) \mathbf{R}_3(-\omega) \mathbf{r}_{\text{perif.}}; \quad \mathbf{r}_{\text{perif.}} = [r \cos \nu, r \sin \nu, 0]^\top \quad (1)$$

with $r = a(1 - e^2)/(1 + e \cos \nu)$. Neglecting Earth rotation introduces $< 2\text{km}$ error-negligible for atmospheric density forecasting.

Every sample is converted to $\mathbf{X} \in \mathbb{R}^{1440 \times 181}$: 168 dynamic OMNI2 features, six raw orbital elements, six sinusoidal encodings (longitude, day-of-year, sidereal time; each as sin/cos), and a file-level prior which normalizes altitude-dependent trends. Histories shorter than 60 days are padded by repeating the last valid row, whereas longer histories are trimmed to the most recent 1440h.

A two-stage scaler is applied: (i) a quantile transformer maps each feature to an approximately Gaussian distribution, mitigating the heavy tails of proton flux and K_p ; (ii) a standard scaler zero-centers and unit-scales the log-residual labels

$$y = \log_{10}(\rho/\rho_0). \quad (2)$$

$$\rho_0 = \mathbb{E}[\rho \mid \rho < 1 \text{ kg m}^{-3}] \quad (3)$$

Both transformers are fitted exclusively on the training partition and saved alongside the network weights, ensuring bit-wise reproducibility.

Methods

The network (Table 1) consumes a 60-day history tensor at 1h cadence—1440 time-steps with 181 scalar features each—and emits a 432-step forecast of orbit-averaged density at 10min resolution (3 days).

A three-layer BiGRU with hidden width $h = 384$ per direction encodes the sequence, yielding hidden states $\mathbf{H} \in \mathbb{R}^{1440 \times 768}$. Additive attention then assigns a probability mass, α_t , to each timestep

$$e_t = \mathbf{w}^\top \tanh(\mathbf{W}\mathbf{h}_t + \mathbf{b}) \quad (4)$$

$$\alpha_t = \text{softmax}(e_t) \quad (5)$$

and forms a context vector

$$\mathbf{z} = \sum_{t=1}^{1440} \alpha_t \mathbf{h}_t, \quad (6)$$

which typically concentrates weight on storm-informative hours. A LayerNorm-GELU-Dropout block projects $\mathbf{z}$ to 384 dimensions, and a final linear layer produces the log-residual trajectory $\hat{\mathbf{y}} \in \mathbb{R}^{432}$.

Training minimizes a time-weighted meansquared error identical in form to the competition metric:

$$\mathcal{L} = \frac{1}{S} \sum_{s=1}^S w_s (\hat{y}_s - y_s)^2, \quad (7)$$

$$w_s = e^{-\gamma s \Delta t}; \quad \gamma = -\frac{\ln(\varepsilon)}{S \Delta t} \quad (8)$$

where $S = 432$ is the 3day horizon, $\Delta t = 600\text{s}$ is the 10min cadence, and $\varepsilon = 10^{-5}$ sets the terminal weight. Because $\sum_{s=1}^k w_s / \sum_{s=1}^S w_s = 1 - \varepsilon^{k/S}$, 95% of the total mass lies in the first $k \approx 0.26 S$ steps ($\sim 19\text{h}$) and 99% within $\sim 29\text{h}$, faithfully mirroring the OD-RMSE weighting (Equation 9).

Table 1: BiGRU-Attention network for 60-day density forecasting. Shapes follow (timesteps, features).

Stage	Layer	Output shape	Notes
Encoder			
Input	QC history	(1440, 181)	168 OMNI features +13 static
1–3	3×BiGRU ($h = 384$)	(1440, 768)	Bidirectional, dropout 0.30
4	Additive attention ((6))	(768)	Context vector $\mathbf{z}$
Predictor MLP			
5	LayerNorm + Linear	(384)	
6	GELU + Dropout ($p = 0.30$)	(384)	
7	Linear	(432)	3 days @ 10min

The model uses AdamW ($\eta_0 = 3 \times 10^{-4}$, weight-decay 10^{-4}), gradient-norm clipping to 1.0, and a Reduce-LROn-Plateau scheduler (factor 0.5, patience 4). Automatic mixed precision (AMP) and a fixed random seed 42 ensure bit-wise reproducibility; PyTorch’s deterministic algorithms are enabled when available. Early stopping with a patience of 24 epochs typically converges in ~ 4 h on a single RTX 3090 Ti. Models are evaluated with identical time-weighted RMSE on an 80/20 (split by *File ID*) validation protocol, ensuring that there is no orbital overlap between the train and test examples.

Results

Leaderboard rank is determined by the skill formulation of the time-weighted root-mean-square error (OD-RMSE)

$$\text{OD-RMSE} = \frac{\sum_t w_t (1 - \text{RMSE}_{\text{test},t} / \text{RMSE}_{\text{MSIS},t})}{\sum_t w_t}, \quad (9)$$

$$w_t = e^{-\gamma t}, \quad \gamma = -\frac{\ln(\epsilon)}{T}, \quad (10)$$

where t is the offset (s) from the start of the evaluation window, T the window length, ϵ is a small value (10^{-5}) that sets the minimum weight, and γ ensures $w_t \approx 10^{-5}$ at $t = T$. Values range from $-\infty$ to 1, where 1 indicates perfect performance and values ≤ 0 indicate no improvement or worse than the baseline. Consequently, $\approx 95\%$ of the weight lies in the first 48h of each 72h forecast.

On the Phase 1 Medium and Hard datasets, the proposed BiGRU-Attention model attains a composite score of 0.709, ranking 5th out of 18 validated submissions (Table 2) and surpassing the public transformer baseline.

Table 2: Official Phase 1 results (NRLMSIS baseline score = 0).

Phase 1.1 Public (Medium)	Phase 1.1 Public (Hard)	Public Score	Private Score	Model Score, M	Normalised Score, M_{norm} $\uparrow$
0.678400	0.539800	0.567543	0.115586	0.499750	0.709428

Figure 1 tracks RMSE over 55 training epochs. The training error (blue) decays smoothly, while the validation curve (orange) plateaus after epoch ~ 20 , indicating that the early-stop trigger prevents over-fitting, and the model converges in fewer than 30 epochs under the prescribed learning rate schedule.

Figure 2 reports error as a function of lead time. The unweighted RMSE (blue) remains nearly constant across the 72h horizon, evidencing stable point accuracy. In contrast, the OD-weighted RMSE (orange) decreases exponentially, mirroring the scoring weights, and falls two orders of magnitude below the baseline

by 72h. This confirms that the network captures the rapid decay of space-weather influence on orbit-mean density and allocates model capacity where it matters most for competition scoring.

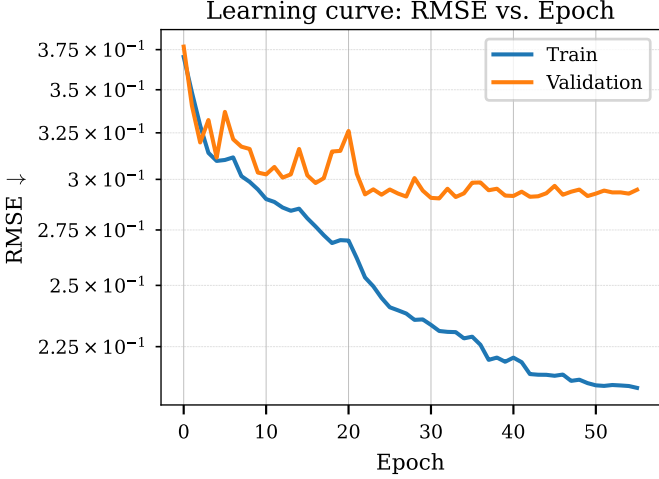


Figure 1: Evolution of train and validation RMSE over epochs

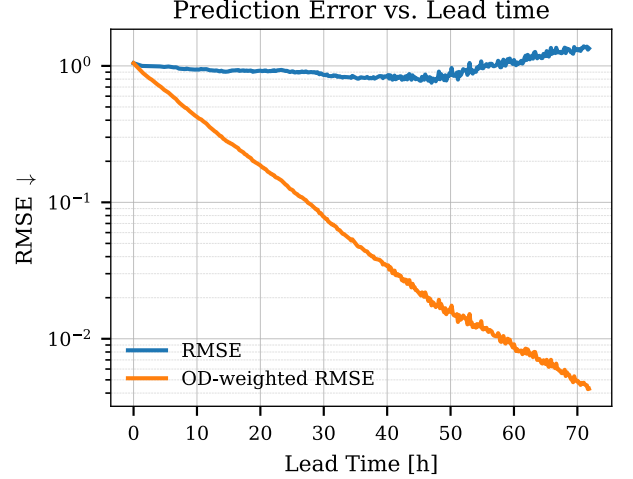


Figure 2: Prediction error versus lead time. The OD-weighted curve reflects the exponential emphasis on early horizons used for leaderboard scoring.

Discussion & Model Evolution

Despite its real-time throughput and leaderboard performance, the purely empirical BiGRU-Attention model is constrained by the statistics of its training corpus; rare storm morphologies or long-scale composition changes may be extrapolated poorly. A natural extension is a *hybrid* scheme that adds the learned residual to a physics prior such as MSIS 2.1, enabling out-of-distribution robustness and continuous bias correction through data assimilation [6]. Skill could be further bolstered by augmenting the OMNI2 features with high-cadence EUV flux, composition ratios, or real-time F10.7 proxies, routed through modality-specific encoders to sharpen storm-onset timing [7].

The current model is static and provides point estimates only. Lightweight continual-learning updates-periodic fine-tuning or rehearsal buffers-would guard against secular drifts in solar activity. At the same time, Bayesian last-layer Laplace or Monte-Carlo dropout could supply calibrated uncertainty bands essential for collision-risk analysis. Together, these refinements would evolve the prototype into a resilient operational forecaster without sacrificing its (sub)millisecond inference speed.

Conclusion & Broader Impact

We introduced a fully data-driven BiGRU-Attention network that converts 60 days of solar wind history and orbital context into a 72h forecast of orbit-averaged density, achieving a composite OD-RMSE skill of 0.709 on the STORM-AI Phase 1 leaderboard, surpassing transformer baselines while retaining millisecond inference. Metric-aligned loss weighting, per-sequence normalization, and multiscale lag features jointly support the model’s storm-time fidelity and cross-satellite generalization, and the architecture is sufficiently lightweight for deployment on flight-dynamics servers or even onboard processors to deliver real-time drag estimates for maneuver planning and conjunction screening.

Future extensions that fuse physics priors, ingest higher-cadence space weather streams, incorporate continual learning, and output calibrated uncertainties will transform this prototype into a robust operational tool, thereby enhancing situational awareness for the rapidly expanding LEO population and mitigating collision risk amid increasing space-weather volatility.

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